

Can Infinity Collapse to Zero?

A Universal Infinite-Digit Representation System

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Abstract

We explore a universal representation for real numbers using infinite digits on both sides of the decimal point. This framework merges properties of real analysis [7] and b -adic number theory [5]. In this system, the symmetric maximal-digit expansion

$$\dots (b-1)(b-1)(b-1) \cdot (b-1)(b-1)(b-1) \dots$$

evaluates to 0, leading to the provocative question of whether the “largest” representable number collapses to zero. We show how negative numbers can be represented without an explicit minus sign, demonstrate generalized zero patterns, and unify left-infinite (b -adic) and right-infinite (real-analytic) interpretations.

1 Introduction

Traditional real numbers allow infinite expansion only to the right of the decimal point [4], while the left side remains finite. We propose a universal infinite-digit format:

$$\dots d_3 d_2 d_1 d_0 \cdot d'_0 d'_1 d'_2 \dots$$

This merges real decimal expansions with b -adic expansions [3]. We explore whether symmetric infinite expansions allow infinity and zero to coincide. A general numeration context is given in [2]. The broad philosophical motivation aligns with ideas in [1].

2 Standard Real Representations

Real numbers use finite left expansions and possibly infinite right expansions. Negative numbers require an explicit minus sign. Classical decimal structure is reviewed in [4].

3 Extending the Representation

We define the general infinite-digit representation and show how left-infinite digits behave analogously to b -adic integers while right-infinite digits behave like real limits.

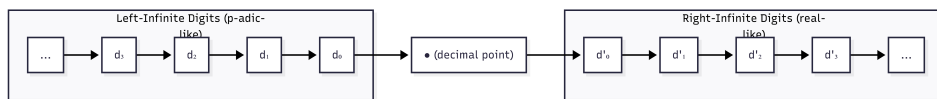


Figure 1: Universal infinite-digit representation

4 Why Infinite Left Digits Are Necessary

In standard positional notation, every real number requires only finitely many digits to the left of the decimal point. Even extremely large values such as

$$10^{1000} = 1 \underbrace{00 \dots 0}_{1000 \text{ zeros}}$$

still involve a finite left block.

However, this finiteness introduces a structural limitation. If we restrict a representation system to at most n left digits, then the largest number that can be expressed has the form

$$b^n - 1.$$

But for any such finite n , we can always construct a strictly larger number:

$$b^n,$$

which requires $n + 1$ digits to represent.

Because the real numbers are unbounded above, no fixed value of n can accommodate all reals using only n left digits. Therefore:

A number representation system capable of encoding *all* real numbers must allow arbitrarily many digits to the left of the decimal point.

The simplest consistent solution is to adopt a left-infinite structure:

$$\dots d_3 d_2 d_1 d_0 . d'_0 d'_1 d'_2 \dots$$

This perfectly mirrors the right-infinite structure used for fractional precision in real analysis, providing symmetry between magnitude and precision.

5 Infinite Left-Digit Expansions

$$(b^1 - 1), (b^2 - 1), (b^3 - 1), \dots$$

In the b -adic metric, $b^n \rightarrow 0$, so $b^n - 1 \rightarrow -1$ [5].

$$\dots (b-1)(b-1)(b-1) = -1.$$

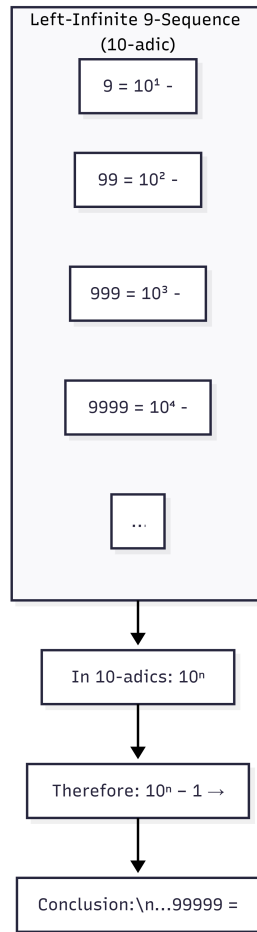


Figure 2: Limit-based interpretation of $\dots (b-1)(b-1)(b-1)$ as -1

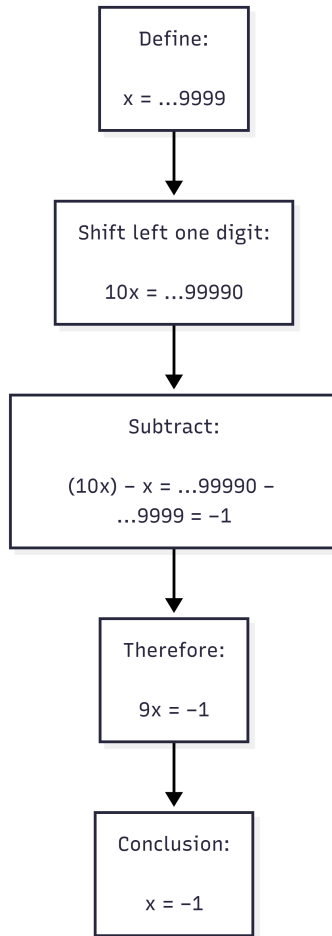


Figure 3: Algebraic shift proof that $\dots(b-1)(b-1)(b-1) = -1$

6 Why the Right Side Equals +1

$$\dots(b-1)(b-1)(b-1)\dots = \sum_{n=1}^{\infty} \frac{b-1}{b^n} = 1.$$

[7, 6]

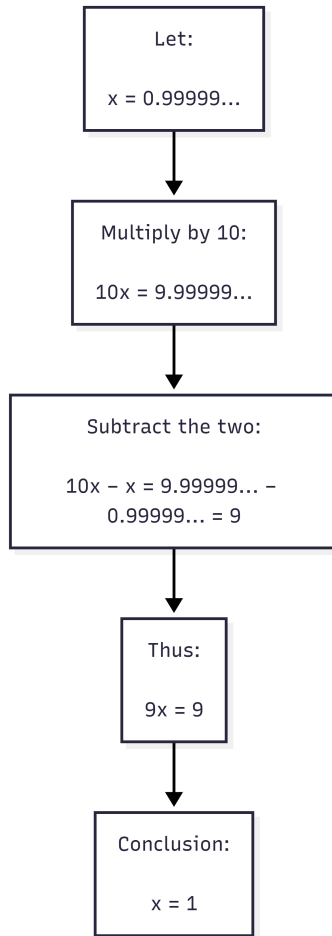


Figure 4: Algebraic proof that $0.999\dots = 1$

7 Symmetric Combination: The Collapse to Zero

$$(-1) + (1) = 0.$$

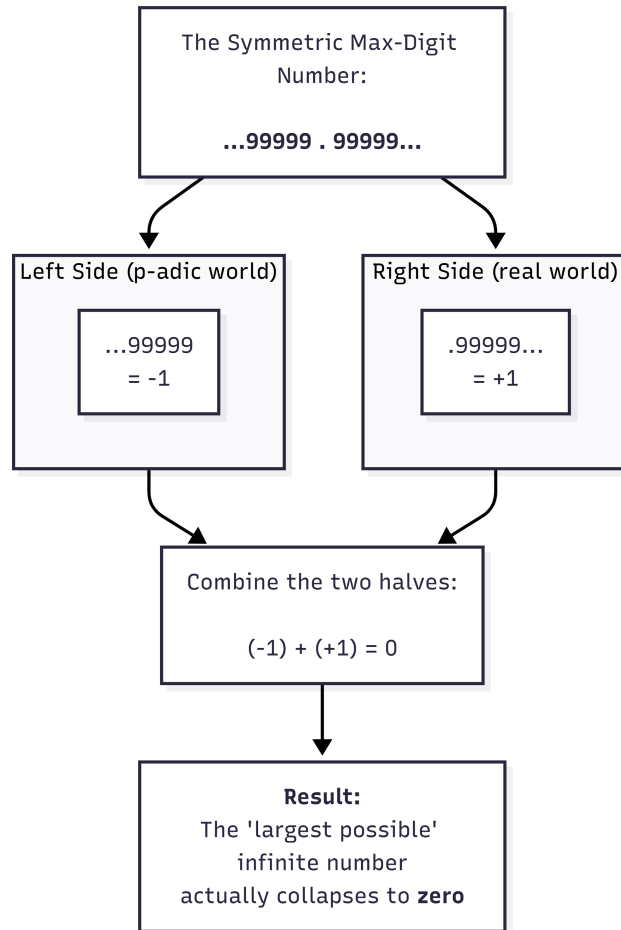


Figure 5: The largest symmetric expansion collapses to zero

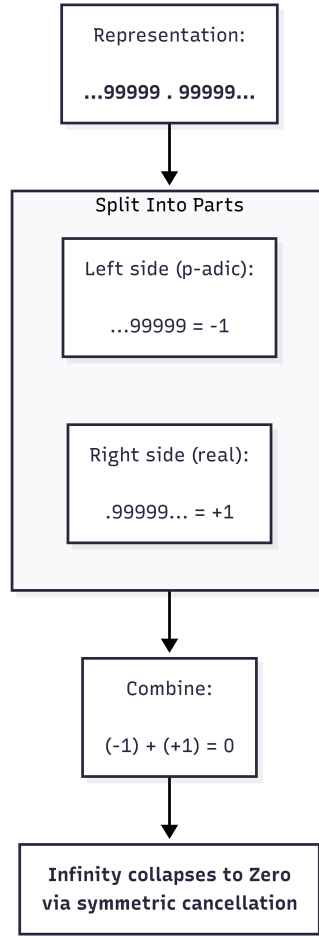


Figure 6: Cancellation of left-infinite and right-infinite contributions

8 General Constant-Digit Zero Patterns

$$\dots dddd.dddd\dots = 0$$

$$d = \frac{d}{b-1}(b-1)$$

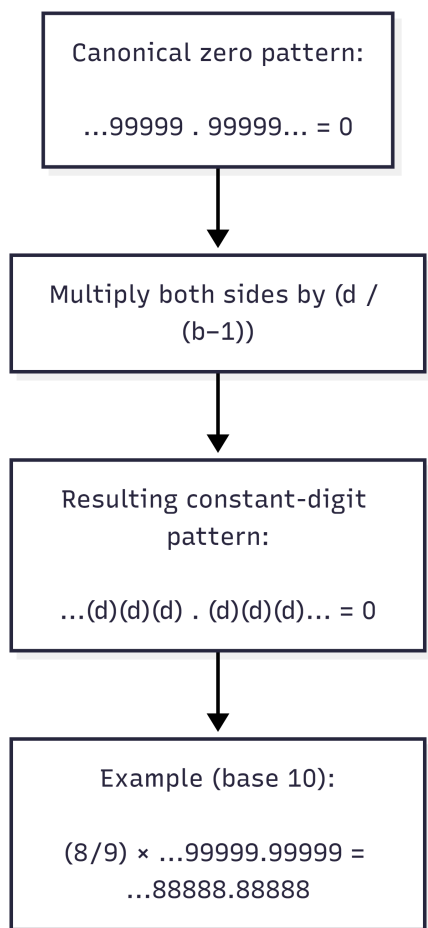


Figure 7: All constant-digit symmetric patterns represent zero

9 Negative Numbers Without a Minus Sign

$$-1 = \dots 1110.1111\dots$$

$$-37 = \dots 666629.66666\dots$$

$$-37 = \dots 444407.444444\dots$$

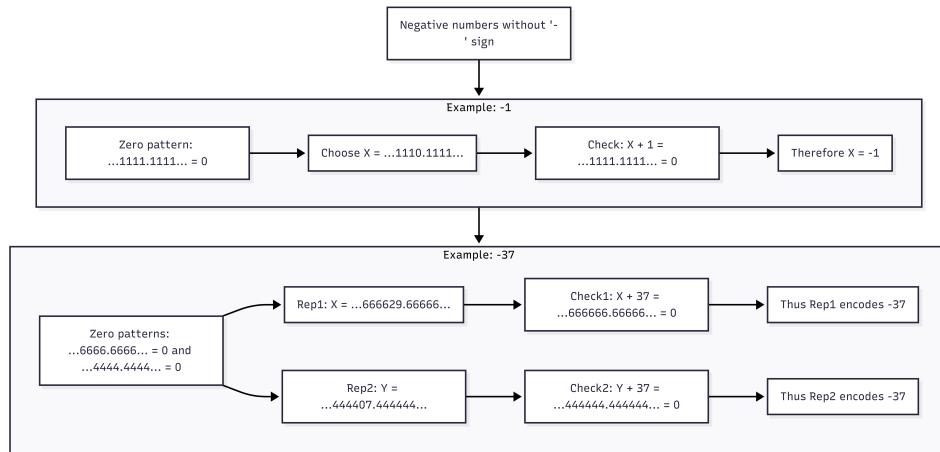


Figure 8: Representing negative values without a minus sign

10 Invalid Patterns: Why 9999999... Is Undefined

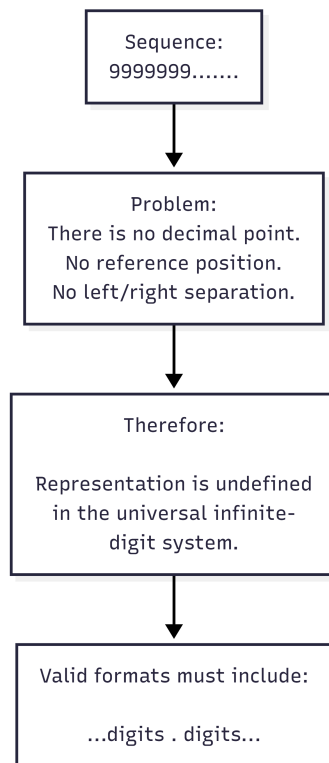


Figure 9: Why patterns without a decimal point are undefined

11 Unified Architecture

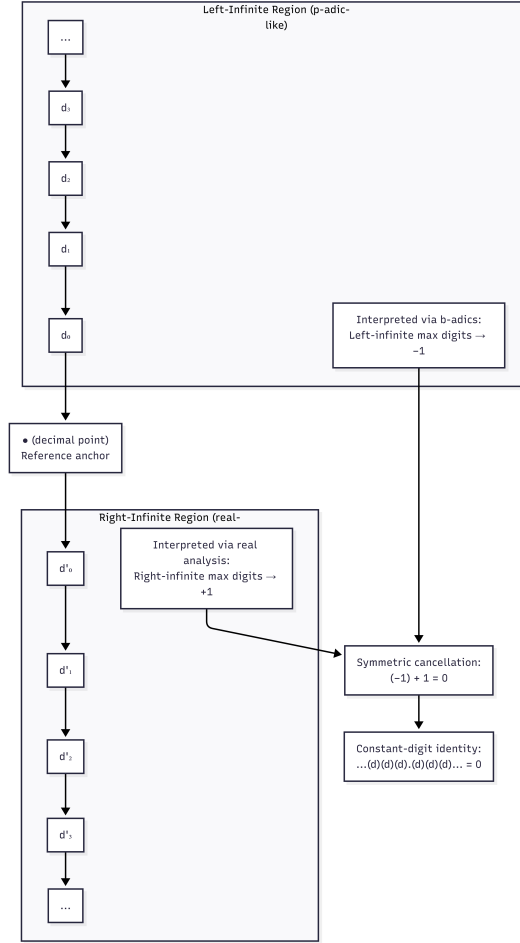


Figure 10: Unified architecture of the infinite-digit representation system

12 Conclusion

We unify real and b -adic systems into one infinite-digit representation. The symmetric collapse phenomenon demonstrates a deep connection between b -adic limits and real analytic limits.

References

- [1] Can infinity collapse to zero? (video inspiration). https://youtu.be/tRaq4aYPzCc?si=88m_NhE_9kM2Vz7C. YouTube reference, accessed 2025.
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- [6] Konrad Knopp. On the representation $0.999\ldots = 1$. *Mathematische Zeitschrift*, 1928. Classic discussion of repeating decimal limits.
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